SG No. 230 O_{h}^{10} $Ia\bar{3}d$ [cubic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$
* Wyckoff site: `16a`, site symmetry: `.-3.`

Table 1: Wyckoff bond: `16a@16a`, bond symmetry: `.-3.`

No.	vector	center	mapping
1	$[X,\ X,\ X]$	$[0,\ 0,\ 0]$	`[1,5,9,-25,-29,-33]`
2	$[-X,\ -X,\ X]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,7,12,-26,-31,-36]`
3	$[-X,\ X,\ -X]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,8,10,-27,-32,-34]`
4	$[X,\ -X,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,6,11,-28,-30,-35]`
5	$[X,\ X,\ -X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13,17,21,-37,-41,-45]`
6	$[-X,\ -X,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[14,19,24,-38,-43,-48]`
7	$[X,\ -X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[15,20,22,-39,-44,-46]`
8	$[-X,\ X,\ X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[16,18,23,-40,-42,-47]`

Table 2: Wyckoff bond: `48b@16a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ 0]$	`[1,-25]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,-26]`
3	$[-X,\ Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,-27]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,-28]`
5	$[Z,\ X,\ Y]$	$[0,\ 0,\ 0]$	`[5,-29]`
6	$[Z,\ -X,\ -Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[6,-30]`
7	$[-Z,\ -X,\ Y]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[7,-31]`
8	$[-Z,\ X,\ -Y]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[8,-32]`
9	$[Y,\ Z,\ X]$	$[0,\ 0,\ 0]$	`[9,-33]`
10	$[-Y,\ Z,\ -X]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,-34]`
11	$[Y,\ -Z,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[11,-35]`
12	$[-Y,\ -Z,\ X]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[12,-36]`
13	$[Y,\ X,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13,-37]`
14	$[-Y,\ -X,\ -Z]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[14,-38]`
15	$[Y,\ -X,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[15,-39]`
16	$[-Y,\ X,\ Z]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[16,-40]`
17	$[X,\ Z,\ -Y]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[17,-41]`
18	$[-X,\ Z,\ Y]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[18,-42]`
19	$[-X,\ -Z,\ -Y]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[19,-43]`
20	$[X,\ -Z,\ Y]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[20,-44]`
21	$[Z,\ Y,\ -X]$	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[21,-45]`
22	$[Z,\ -Y,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[22,-46]`
23	$[-Z,\ Y,\ X]$	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[23,-47]`
24	$[-Z,\ -Y,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[24,-48]`

* Wyckoff site: `16b`, site symmetry: `.32`

Table 3: Wyckoff bond: `16a@16b`, bond symmetry: `.32`

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[1,5,9,-14,-19,-24]`
2	$[-X, -X, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[2,7,12,-13,-17,-21]`
3	$[-X, X, -X]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[3,8,10,-15,-20,-22]`
4	$[X, -X, -X]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[4,6,11,-16,-18,-23]`
5	$[-X, -X, -X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[25,29,33,-38,-43,-48]`
6	$[X, X, -X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[26,31,36,-37,-41,-45]`
7	$[X, -X, X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[27,32,34,-39,-44,-46]`
8	$[-X, X, X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[28,30,35,-40,-42,-47]`

Table 4: Wyckoff bond: `48b@16b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[1,-14]`
2	$[-X, -X, Y]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[2,-13]`
3	$[-X, X, -Y]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[3,-15]`
4	$[X, -X, -Y]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[4,-16]`
5	$[Y, X, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[5,-24]`
6	$[Y, -X, -X]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[6,-23]`
7	$[-Y, -X, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[7,-21]`
8	$[-Y, X, -X]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[8,-22]`
9	$[X, Y, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[9,-19]`
10	$[-X, Y, -X]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[10,-20]`
11	$[X, -Y, -X]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[11,-18]`
12	$[-X, -Y, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[12,-17]`
13	$[-X, -X, -Y]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[25,-38]`
14	$[X, X, -Y]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[26,-37]`
15	$[X, -X, Y]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[27,-39]`
16	$[-X, X, Y]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[28,-40]`
17	$[-Y, -X, -X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[29,-48]`
18	$[-Y, X, X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[30,-47]`
19	$[Y, X, -X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[31,-45]`
20	$[Y, -X, X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[32,-46]`
21	$[-X, -Y, -X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[33,-43]`
22	$[X, -Y, X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[34,-44]`
23	$[-X, Y, X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[35,-42]`
24	$[X, Y, -X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[36,-41]`

Table 5: Wyckoff bond: `48c@16b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[1,14]`
2	$[-X, X, 0]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[2,13]`
3	$[-X, -X, 0]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[3,15]`
4	$[X, X, 0]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[4,16]`

continued ...

Table 5

No.	vector	center	mapping
5	$[0, X, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[5,24]
6	$[0, -X, X]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	[6,23]
7	$[0, -X, -X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	[7,21]
8	$[0, X, X]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	[8,22]
9	$[-X, 0, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[9,19]
10	$[X, 0, -X]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	[10,20]
11	$[-X, 0, -X]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	[11,18]
12	$[X, 0, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	[12,17]
13	$[-X, X, 0]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	[25,38]
14	$[X, -X, 0]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	[26,37]
15	$[X, X, 0]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	[27,39]
16	$[-X, -X, 0]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	[28,40]
17	$[0, -X, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	[29,48]
18	$[0, X, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	[30,47]
19	$[0, X, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	[31,45]
20	$[0, -X, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	[32,46]
21	$[X, 0, -X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	[33,43]
22	$[-X, 0, X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	[34,44]
23	$[X, 0, X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	[35,42]
24	$[-X, 0, -X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	[36,41]

Table 6: Wyckoff bond: 96d@16b, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[1]
2	$[-X, -Y, Z]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	[2]
3	$[-X, Y, -Z]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	[3]
4	$[X, -Y, -Z]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	[4]
5	$[Z, X, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[5]
6	$[Z, -X, -Y]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	[6]
7	$[-Z, -X, Y]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	[7]
8	$[-Z, X, -Y]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	[8]
9	$[Y, Z, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[9]
10	$[-Y, Z, -X]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	[10]
11	$[Y, -Z, -X]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	[11]
12	$[-Y, -Z, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	[12]
13	$[Y, X, -Z]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	[13]
14	$[-Y, -X, -Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[14]
15	$[Y, -X, Z]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	[15]
16	$[-Y, X, Z]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	[16]
17	$[X, Z, -Y]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	[17]
18	$[-X, Z, Y]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	[18]
19	$[-X, -Z, -Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[19]
20	$[X, -Z, Y]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	[20]
21	$[Z, Y, -X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	[21]

continued ...

Table 6

No.	vector	center	mapping
22	$[Z, -Y, X]$	$\left[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[25]
26	$[X, Y, -Z]$	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[26]
27	$[X, -Y, Z]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[48]

* Wyckoff site: 24c, site symmetry: 2.22

Table 7: Wyckoff bond: 24a@24c, bond symmetry: 2.22

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	[1,-4,18,-19]
2	$[0, -X, X]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	[2,-3,-17,20]
3	$[X, 0, X]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	[5,-8,-14,15]
4	$[X, 0, -X]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	[6,-7,13,-16]
5	$[X, X, 0]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	[9,-12,21,-24]
6	$[-X, X, 0]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	[10,-11,-22,23]
7	$[0, -X, -X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	[25,-28,42,-43]
8	$[0, X, -X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	[26,-27,-41,44]
9	$[-X, 0, -X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	[29,-32,-38,39]
10	$[-X, 0, X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	[30,-31,37,-40]
11	$[-X, -X, 0]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	[33,-36,45,-48]
12	$[X, -X, 0]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	[34,-35,-46,47]

Table 8: Wyckoff bond: `24b@24c`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0,\ X,\ -X]$	$\left[\frac{1}{8},\ 0,\ \frac{1}{4}\right]$	`[1,-4,-18,19]`
2	$[0,\ -X,\ -X]$	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[2,-3,17,-20]`
3	$[-X,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[5,-8,14,-15]`
4	$[-X,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[6,-7,-13,16]`
5	$[X,\ -X,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[9,-12,-21,24]`
6	$[-X,\ -X,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{3}{8}\right]$	`[10,-11,22,-23]`
7	$[0,\ -X,\ X]$	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[25,-28,-42,43]`
8	$[0,\ X,\ X]$	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[26,-27,41,-44]`
9	$[X,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[29,-32,38,-39]`
10	$[X,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[30,-31,-37,40]`
11	$[-X,\ X,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[33,-36,-45,48]`
12	$[X,\ X,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[34,-35,46,-47]`

Table 9: Wyckoff bond: `24c@24c`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{8},\ 0,\ \frac{1}{4}\right]$	`[1,4,-18,-19]`
2	$[-X,\ 0,\ 0]$	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[2,3,-17,-20]`
3	$[0,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[5,8,-14,-15]`
4	$[0,\ -X,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[6,7,-13,-16]`
5	$[0,\ 0,\ X]$	$\left[0,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[9,12,-21,-24]`
6	$[0,\ 0,\ -X]$	$\left[0,\ \frac{3}{4},\ \frac{3}{8}\right]$	`[10,11,-22,-23]`
7	$[-X,\ 0,\ 0]$	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[25,28,-42,-43]`
8	$[X,\ 0,\ 0]$	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[26,27,-41,-44]`
9	$[0,\ -X,\ 0]$	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[29,32,-38,-39]`
10	$[0,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[30,31,-37,-40]`
11	$[0,\ 0,\ -X]$	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[33,36,-45,-48]`
12	$[0,\ 0,\ X]$	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[34,35,-46,-47]`

Table 10: Wyckoff bond: `48d@24c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y,\ X,\ X]$	$\left[\frac{1}{8},\ 0,\ \frac{1}{4}\right]$	`[1,-19]`
2	$[-Y,\ -X,\ X]$	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[2,-17]`
3	$[-Y,\ X,\ -X]$	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[3,-20]`
4	$[Y,\ -X,\ -X]$	$\left[\frac{1}{8},\ 0,\ \frac{1}{4}\right]$	`[4,-18]`
5	$[X,\ Y,\ X]$	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[5,-14]`
6	$[X,\ -Y,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[6,-16]`
7	$[-X,\ -Y,\ X]$	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[7,-13]`
8	$[-X,\ Y,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[8,-15]`
9	$[X,\ X,\ Y]$	$\left[0,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[9,-24]`
10	$[-X,\ X,\ -Y]$	$\left[0,\ \frac{3}{4},\ \frac{3}{8}\right]$	`[10,-22]`
11	$[X,\ -X,\ -Y]$	$\left[0,\ \frac{3}{4},\ \frac{3}{8}\right]$	`[11,-23]`
12	$[-X,\ -X,\ Y]$	$\left[0,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[12,-21]`

continued ...

Table 10

No.	vector	center	mapping
13	$[-Y, -X, -X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[25,-43]`
14	$[Y, X, -X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[26,-41]`
15	$[Y, -X, X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[27,-44]`
16	$[-Y, X, X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[28,-42]`
17	$[-X, -Y, -X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[29,-38]`
18	$[-X, Y, X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[30,-40]`
19	$[X, Y, -X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[31,-37]`
20	$[X, -Y, X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[32,-39]`
21	$[-X, -X, -Y]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[33,-48]`
22	$[X, -X, Y]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[34,-46]`
23	$[-X, X, Y]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[35,-47]`
24	$[X, X, -Y]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[36,-45]`

Table 11: Wyckoff bond: `48e@24c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, -X]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[1,-18]`
2	$[-Y, -X, -X]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[2,-20]`
3	$[-Y, X, X]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[3,-17]`
4	$[Y, -X, X]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[4,-19]`
5	$[-X, Y, X]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[5,-15]`
6	$[-X, -Y, -X]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[6,-13]`
7	$[X, -Y, X]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[7,-16]`
8	$[X, Y, -X]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[8,-14]`
9	$[X, -X, Y]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[9,-21]`
10	$[-X, -X, -Y]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[10,-23]`
11	$[X, X, -Y]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[11,-22]`
12	$[-X, X, Y]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[12,-24]`
13	$[-Y, -X, X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[25,-42]`
14	$[Y, X, X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[26,-44]`
15	$[Y, -X, -X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[27,-41]`
16	$[-Y, X, -X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[28,-43]`
17	$[X, -Y, -X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[29,-39]`
18	$[X, Y, X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[30,-37]`
19	$[-X, Y, -X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[31,-40]`
20	$[-X, -Y, X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[32,-38]`
21	$[-X, X, -Y]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[33,-45]`
22	$[X, X, Y]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[34,-47]`
23	$[-X, -X, Y]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[35,-46]`
24	$[X, -X, -Y]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[36,-48]`

Table 12: Wyckoff bond: `48f@24c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ X,\ Y]$	$\left[\frac{1}{8},\ 0,\ \frac{1}{4}\right]$	`[1,-4]`
2	$[0,\ -X,\ Y]$	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[2,-3]`
3	$[Y,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[5,-8]`
4	$[Y,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[6,-7]`
5	$[X,\ Y,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[9,-12]`
6	$[-X,\ Y,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{3}{8}\right]$	`[10,-11]`
7	$[X,\ 0,\ -Y]$	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[13,-16]`
8	$[-X,\ 0,\ -Y]$	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[14,-15]`
9	$[0,\ Y,\ -X]$	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[17,-20]`
10	$[0,\ Y,\ X]$	$\left[\frac{1}{8},\ 0,\ \frac{1}{4}\right]$	`[18,-19]`
11	$[Y,\ X,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[21,-24]`
12	$[Y,\ -X,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{3}{8}\right]$	`[22,-23]`
13	$[0,\ -X,\ -Y]$	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[25,-28]`
14	$[0,\ X,\ -Y]$	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[26,-27]`
15	$[-Y,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[29,-32]`
16	$[-Y,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[30,-31]`
17	$[-X,\ -Y,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[33,-36]`
18	$[X,\ -Y,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[34,-35]`
19	$[-X,\ 0,\ Y]$	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[37,-40]`
20	$[X,\ 0,\ Y]$	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[38,-39]`
21	$[0,\ -Y,\ X]$	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[41,-44]`
22	$[0,\ -Y,\ -X]$	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[42,-43]`
23	$[-Y,\ -X,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[45,-48]`
24	$[-Y,\ X,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[46,-47]`

Table 13: Wyckoff bond: `96g@24c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{8},\ 0,\ \frac{1}{4}\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{8},\ 0,\ \frac{1}{4}\right]$	`[4]`
5	$[Z,\ X,\ Y]$	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[6]`
7	$[-Z,\ -X,\ Y]$	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[7]`
8	$[-Z,\ X,\ -Y]$	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[8]`
9	$[Y,\ Z,\ X]$	$\left[0,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[9]`
10	$[-Y,\ Z,\ -X]$	$\left[0,\ \frac{3}{4},\ \frac{3}{8}\right]$	`[10]`
11	$[Y,\ -Z,\ -X]$	$\left[0,\ \frac{3}{4},\ \frac{3}{8}\right]$	`[11]`
12	$[-Y,\ -Z,\ X]$	$\left[0,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[12]`
13	$[Y,\ X,\ -Z]$	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[13]`
14	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[14]`
15	$[Y,\ -X,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[15]`
16	$[-Y,\ X,\ Z]$	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[16]`
17	$[X,\ Z,\ -Y]$	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[17]`

continued ...

Table 13

No.	vector	center	mapping
18	$[-X, Z, Y]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	[19]
20	$[X, -Z, Y]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	[21]
22	$[Z, -Y, X]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	[22]
23	$[-Z, Y, X]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	[25]
26	$[X, Y, -Z]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	[26]
27	$[X, -Y, Z]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	[33]
34	$[Y, -Z, X]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	[34]
35	$[-Y, Z, X]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	[35]
36	$[Y, Z, -X]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	[47]
48	$[Z, Y, X]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	[48]

* Wyckoff site: 24d, site symmetry: -4..

Table 14: Wyckoff bond: 24a@24d, bond symmetry: -4..

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{3}{8}, 0, \frac{1}{4}\right]$	[1,4,-41,-44]
2	$[-X, 0, 0]$	$\left[\frac{1}{8}, 0, \frac{3}{4}\right]$	[2,3,-42,-43]
3	$[0, X, 0]$	$\left[\frac{1}{4}, \frac{3}{8}, 0\right]$	[5,8,-37,-40]
4	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{8}, 0\right]$	[6,7,-38,-39]
5	$[0, 0, X]$	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	[9,12,-46,-47]
6	$[0, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	[10,11,-45,-48]
7	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{5}{8}, 0\right]$	[13,16,-29,-32]
8	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{8}, \frac{1}{2}\right]$	[14,15,-30,-31]
9	$[X, 0, 0]$	$\left[\frac{1}{8}, \frac{1}{2}, \frac{1}{4}\right]$	[17,20,-25,-28]
10	$[-X, 0, 0]$	$\left[\frac{7}{8}, 0, \frac{1}{4}\right]$	[18,19,-26,-27]

continued ...

Table 14

No.	vector	center	mapping
11	$[0, 0, -X]$	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[21,24,-34,-35]`
12	$[0, 0, X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[22,23,-33,-36]`

Table 15: Wyckoff bond: `48b@24d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[\frac{3}{8}, 0, \frac{1}{4}\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[\frac{1}{8}, 0, \frac{3}{4}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[\frac{1}{4}, \frac{3}{8}, 0\right]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{8}, 0\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{3}{4}, \frac{5}{8}, 0\right]$	`[13,-16]`
8	$[-X, 0, -Y]$	$\left[\frac{3}{4}, \frac{3}{8}, \frac{1}{2}\right]$	`[14,-15]`
9	$[0, Y, -X]$	$\left[\frac{1}{8}, \frac{1}{2}, \frac{1}{4}\right]$	`[17,-20]`
10	$[0, Y, X]$	$\left[\frac{7}{8}, 0, \frac{1}{4}\right]$	`[18,-19]`
11	$[Y, X, 0]$	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[21,-24]`
12	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[22,-23]`
13	$[0, -X, -Y]$	$\left[\frac{1}{8}, \frac{1}{2}, \frac{1}{4}\right]$	`[25,-28]`
14	$[0, X, -Y]$	$\left[\frac{7}{8}, 0, \frac{1}{4}\right]$	`[26,-27]`
15	$[-Y, 0, -X]$	$\left[\frac{3}{4}, \frac{5}{8}, 0\right]$	`[29,-32]`
16	$[-Y, 0, X]$	$\left[\frac{3}{4}, \frac{3}{8}, \frac{1}{2}\right]$	`[30,-31]`
17	$[-X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[33,-36]`
18	$[X, -Y, 0]$	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[34,-35]`
19	$[-X, 0, Y]$	$\left[\frac{1}{4}, \frac{3}{8}, 0\right]$	`[37,-40]`
20	$[X, 0, Y]$	$\left[\frac{3}{4}, \frac{1}{8}, 0\right]$	`[38,-39]`
21	$[0, -Y, X]$	$\left[\frac{3}{8}, 0, \frac{1}{4}\right]$	`[41,-44]`
22	$[0, -Y, -X]$	$\left[\frac{1}{8}, 0, \frac{3}{4}\right]$	`[42,-43]`
23	$[-Y, -X, 0]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[45,-48]`
24	$[-Y, X, 0]$	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[46,-47]`

Table 16: Wyckoff bond: `96c@24d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{3}{8}, 0, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{8}, 0, \frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{8}, 0, \frac{3}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{3}{8}, 0, \frac{1}{4}\right]$	`[4]`
5	$[Z, X, Y]$	$\left[\frac{1}{4}, \frac{3}{8}, 0\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[\frac{3}{4}, \frac{1}{8}, 0\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{3}{4}, \frac{1}{8}, 0\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, \frac{3}{8}, 0\right]$	`[8]`
9	$[Y, Z, X]$	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[9]`
10	$[-Y, Z, -X]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[10]`

continued ...

Table 16

No.	vector	center	mapping
11	$[Y, -Z, -X]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, \frac{5}{8}, 0\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{8}, \frac{1}{2}\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[\frac{3}{4}, \frac{3}{8}, \frac{1}{2}\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[\frac{3}{4}, \frac{5}{8}, 0\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[\frac{1}{8}, \frac{1}{2}, \frac{1}{4}\right]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{7}{8}, 0, \frac{1}{4}\right]$	`[18]`
19	$[-X, -Z, -Y]$	$\left[\frac{7}{8}, 0, \frac{1}{4}\right]$	`[19]`
20	$[X, -Z, Y]$	$\left[\frac{1}{8}, \frac{1}{2}, \frac{1}{4}\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[21]`
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[22]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[23]`
24	$[-Z, -Y, -X]$	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[24]`
25	$[-X, -Y, -Z]$	$\left[\frac{1}{8}, \frac{1}{2}, \frac{1}{4}\right]$	`[25]`
26	$[X, Y, -Z]$	$\left[\frac{7}{8}, 0, \frac{1}{4}\right]$	`[26]`
27	$[X, -Y, Z]$	$\left[\frac{7}{8}, 0, \frac{1}{4}\right]$	`[27]`
28	$[-X, Y, Z]$	$\left[\frac{1}{8}, \frac{1}{2}, \frac{1}{4}\right]$	`[28]`
29	$[-Z, -X, -Y]$	$\left[\frac{3}{4}, \frac{5}{8}, 0\right]$	`[29]`
30	$[-Z, X, Y]$	$\left[\frac{3}{4}, \frac{3}{8}, \frac{1}{2}\right]$	`[30]`
31	$[Z, X, -Y]$	$\left[\frac{3}{4}, \frac{3}{8}, \frac{1}{2}\right]$	`[31]`
32	$[Z, -X, Y]$	$\left[\frac{3}{4}, \frac{5}{8}, 0\right]$	`[32]`
33	$[-Y, -Z, -X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[33]`
34	$[Y, -Z, X]$	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[34]`
35	$[-Y, Z, X]$	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[35]`
36	$[Y, Z, -X]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[36]`
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{8}, 0\right]$	`[37]`
38	$[Y, X, Z]$	$\left[\frac{3}{4}, \frac{1}{8}, 0\right]$	`[38]`
39	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{8}, 0\right]$	`[39]`
40	$[Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{3}{8}, 0\right]$	`[40]`
41	$[-X, -Z, Y]$	$\left[\frac{3}{8}, 0, \frac{1}{4}\right]$	`[41]`
42	$[X, -Z, -Y]$	$\left[\frac{1}{8}, 0, \frac{3}{4}\right]$	`[42]`
43	$[X, Z, Y]$	$\left[\frac{1}{8}, 0, \frac{3}{4}\right]$	`[43]`
44	$[-X, Z, -Y]$	$\left[\frac{3}{8}, 0, \frac{1}{4}\right]$	`[44]`
45	$[-Z, -Y, X]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[45]`
46	$[-Z, Y, -X]$	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[46]`
47	$[Z, -Y, -X]$	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[47]`
48	$[Z, Y, X]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[48]`

* Wyckoff site: `32e`, site symmetry: `.3.`

Table 17: Wyckoff bond: `32a@32e`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	`[1,5,9]`
2	$[-X, -X, X]$	$\left[\frac{1}{2} - x, -x, x + \frac{1}{2}\right]$	`[2,7,12]`
3	$[-X, X, -X]$	$\left[-x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	`[3,8,10]`

continued ...

Table 17

No.	vector	center	mapping
4	$[X, -X, -X]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, -x\right]$	[4,6,11]
5	$[X, X, -X]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{4}-x\right]$	[13,17,21]
6	$[-X, -X, -X]$	$\left[\frac{3}{4}-x, \frac{3}{4}-x, \frac{3}{4}-x\right]$	[14,19,24]
7	$[X, -X, X]$	$\left[x+\frac{1}{4}, \frac{1}{4}-x, x+\frac{3}{4}\right]$	[15,20,22]
8	$[-X, X, X]$	$\left[\frac{1}{4}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[16,18,23]
9	$[-X, -X, -X]$	$[-x, -x, -x]$	[25,29,33]
10	$[X, X, -X]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}-x\right]$	[26,31,36]
11	$[X, -X, X]$	$\left[x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[27,32,34]
12	$[-X, X, X]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x\right]$	[28,30,35]
13	$[-X, -X, X]$	$\left[\frac{1}{4}-x, \frac{3}{4}-x, x+\frac{3}{4}\right]$	[37,41,45]
14	$[X, X, X]$	$\left[x+\frac{1}{4}, x+\frac{1}{4}, x+\frac{1}{4}\right]$	[38,43,48]
15	$[-X, X, -X]$	$\left[\frac{3}{4}-x, x+\frac{3}{4}, \frac{1}{4}-x\right]$	[39,44,46]
16	$[X, -X, -X]$	$\left[x+\frac{3}{4}, \frac{1}{4}-x, \frac{3}{4}-x\right]$	[40,42,47]

Table 18: Wyckoff bond: 96b@32e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, -x, x+\frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, -x\right]$	[4]
5	$[Z, X, Y]$	$[x, x, x]$	[5]
6	$[Z, -X, -Y]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, -x\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}-x, -x, x+\frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[8]
9	$[Y, Z, X]$	$[x, x, x]$	[9]
10	$[-Y, Z, -X]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[x+\frac{1}{2}, \frac{1}{2}-x, -x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}-x, -x, x+\frac{1}{2}\right]$	[12]
13	$[Y, X, -Z]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{4}-x\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}-x, \frac{3}{4}-x, \frac{3}{4}-x\right]$	[14]
15	$[Y, -X, Z]$	$\left[x+\frac{1}{4}, \frac{1}{4}-x, x+\frac{3}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{4}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{4}-x\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{4}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{3}{4}-x, \frac{3}{4}-x, \frac{3}{4}-x\right]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{1}{4}, \frac{1}{4}-x, x+\frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{4}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[x+\frac{1}{4}, \frac{1}{4}-x, x+\frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{4}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{3}{4}-x, \frac{3}{4}-x, \frac{3}{4}-x\right]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -x, -x]$	[25]
26	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}-x\right]$	[26]
27	$[X, -Y, Z]$	$\left[x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x\right]$	[28]

continued ...

Table 18

No.	vector	center	mapping
29	$[-Z, -X, -Y]$	$[-x, -x, -x]$	`[29]`
30	$[-Z, X, Y]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, x\right]$	`[30]`
31	$[Z, X, -Y]$	$\left[x + \frac{1}{2}, x, \frac{1}{2} - x\right]$	`[31]`
32	$[Z, -X, Y]$	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	`[32]`
33	$[-Y, -Z, -X]$	$[-x, -x, -x]$	`[33]`
34	$[Y, -Z, X]$	$\left[x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	`[34]`
35	$[-Y, Z, X]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, x\right]$	`[35]`
36	$[Y, Z, -X]$	$\left[x + \frac{1}{2}, x, \frac{1}{2} - x\right]$	`[36]`
37	$[-Y, -X, Z]$	$\left[\frac{1}{4} - x, \frac{3}{4} - x, x + \frac{3}{4}\right]$	`[37]`
38	$[Y, X, Z]$	$\left[x + \frac{1}{4}, x + \frac{1}{4}, x + \frac{1}{4}\right]$	`[38]`
39	$[-Y, X, -Z]$	$\left[\frac{3}{4} - x, x + \frac{3}{4}, \frac{1}{4} - x\right]$	`[39]`
40	$[Y, -X, -Z]$	$\left[x + \frac{3}{4}, \frac{1}{4} - x, \frac{3}{4} - x\right]$	`[40]`
41	$[-X, -Z, Y]$	$\left[\frac{1}{4} - x, \frac{3}{4} - x, x + \frac{3}{4}\right]$	`[41]`
42	$[X, -Z, -Y]$	$\left[x + \frac{3}{4}, \frac{1}{4} - x, \frac{3}{4} - x\right]$	`[42]`
43	$[X, Z, Y]$	$\left[x + \frac{1}{4}, x + \frac{1}{4}, x + \frac{1}{4}\right]$	`[43]`
44	$[-X, Z, -Y]$	$\left[\frac{3}{4} - x, x + \frac{3}{4}, \frac{1}{4} - x\right]$	`[44]`
45	$[-Z, -Y, X]$	$\left[\frac{1}{4} - x, \frac{3}{4} - x, x + \frac{3}{4}\right]$	`[45]`
46	$[-Z, Y, -X]$	$\left[\frac{3}{4} - x, x + \frac{3}{4}, \frac{1}{4} - x\right]$	`[46]`
47	$[Z, -Y, -X]$	$\left[x + \frac{3}{4}, \frac{1}{4} - x, \frac{3}{4} - x\right]$	`[47]`
48	$[Z, Y, X]$	$\left[x + \frac{1}{4}, x + \frac{1}{4}, x + \frac{1}{4}\right]$	`[48]`

* Wyckoff site: `48f`, site symmetry: `2..`

Table 19: Wyckoff bond: `48a@48f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[\frac{1}{2} - x, 0, \frac{3}{4}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[\frac{1}{4}, x, 0\right]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{2} - x, 0\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[0, \frac{1}{4}, x\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[0, \frac{3}{4}, \frac{1}{2} - x\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{3}{4}, x + \frac{1}{4}, 0\right]$	`[13,-16]`
8	$[-X, 0, -Y]$	$\left[\frac{3}{4}, \frac{3}{4} - x, \frac{1}{2}\right]$	`[14,-15]`
9	$[0, Y, -X]$	$\left[x + \frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[17,-20]`
10	$[0, Y, X]$	$\left[\frac{1}{4} - x, 0, \frac{1}{4}\right]$	`[18,-19]`
11	$[Y, X, 0]$	$\left[0, \frac{1}{4}, \frac{1}{4} - x\right]$	`[21,-24]`
12	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, x + \frac{3}{4}\right]$	`[22,-23]`
13	$[0, -X, -Y]$	$\left[-x, 0, \frac{3}{4}\right]$	`[25,-28]`
14	$[0, X, -Y]$	$\left[x + \frac{1}{2}, 0, \frac{1}{4}\right]$	`[26,-27]`
15	$[-Y, 0, -X]$	$\left[\frac{3}{4}, -x, 0\right]$	`[29,-32]`
16	$[-Y, 0, X]$	$\left[\frac{1}{4}, x + \frac{1}{2}, 0\right]$	`[30,-31]`
17	$[-X, -Y, 0]$	$\left[0, \frac{3}{4}, -x\right]$	`[33,-36]`
18	$[X, -Y, 0]$	$\left[0, \frac{1}{4}, x + \frac{1}{2}\right]$	`[34,-35]`
19	$[-X, 0, Y]$	$\left[\frac{1}{4}, \frac{3}{4} - x, 0\right]$	`[37,-40]`
20	$[X, 0, Y]$	$\left[\frac{1}{4}, x + \frac{1}{4}, \frac{1}{2}\right]$	`[38,-39]`
21	$[0, -Y, X]$	$\left[\frac{1}{4} - x, \frac{1}{2}, \frac{3}{4}\right]$	`[41,-44]`

continued ...

Table 19

No.	vector	center	mapping
22	$[0, -Y, -X]$	$\left[x+\frac{3}{4}, 0, \frac{3}{4}\right]$	`[42,-43]`
23	$[-Y, -X, 0]$	$\left[0, \frac{3}{4}, x+\frac{3}{4}\right]$	`[45,-48]`
24	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-x\right]$	`[46,-47]`

Table 20: Wyckoff bond: `48b@48f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, 0, \frac{3}{4}\right]$	`[2,3]`
3	$[0, X, 0]$	$\left[\frac{1}{4}, x, 0\right]$	`[5,8]`
4	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{2}-x, 0\right]$	`[6,7]`
5	$[0, 0, X]$	$\left[0, \frac{1}{4}, x\right]$	`[9,12]`
6	$[0, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}-x\right]$	`[10,11]`
7	$[0, X, 0]$	$\left[\frac{3}{4}, x+\frac{1}{4}, 0\right]$	`[13,16]`
8	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}-x, \frac{1}{2}\right]$	`[14,15]`
9	$[X, 0, 0]$	$\left[x+\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[17,20]`
10	$[-X, 0, 0]$	$\left[\frac{1}{4}-x, 0, \frac{1}{4}\right]$	`[18,19]`
11	$[0, 0, -X]$	$\left[0, \frac{1}{4}, \frac{1}{4}-x\right]$	`[21,24]`
12	$[0, 0, X]$	$\left[\frac{1}{2}, \frac{1}{4}, x+\frac{3}{4}\right]$	`[22,23]`
13	$[-X, 0, 0]$	$\left[-x, 0, \frac{3}{4}\right]$	`[25,28]`
14	$[X, 0, 0]$	$\left[x+\frac{1}{2}, 0, \frac{1}{4}\right]$	`[26,27]`
15	$[0, -X, 0]$	$\left[\frac{3}{4}, -x, 0\right]$	`[29,32]`
16	$[0, X, 0]$	$\left[\frac{1}{4}, x+\frac{1}{2}, 0\right]$	`[30,31]`
17	$[0, 0, -X]$	$\left[0, \frac{3}{4}, -x\right]$	`[33,36]`
18	$[0, 0, X]$	$\left[0, \frac{1}{4}, x+\frac{1}{2}\right]$	`[34,35]`
19	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}-x, 0\right]$	`[37,40]`
20	$[0, X, 0]$	$\left[\frac{1}{4}, x+\frac{1}{4}, \frac{1}{2}\right]$	`[38,39]`
21	$[-X, 0, 0]$	$\left[\frac{1}{4}-x, \frac{1}{2}, \frac{3}{4}\right]$	`[41,44]`
22	$[X, 0, 0]$	$\left[x+\frac{3}{4}, 0, \frac{3}{4}\right]$	`[42,43]`
23	$[0, 0, X]$	$\left[0, \frac{3}{4}, x+\frac{3}{4}\right]$	`[45,48]`
24	$[0, 0, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-x\right]$	`[46,47]`

Table 21: Wyckoff bond: `96c@48f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, 0, \frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, 0, \frac{3}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[4]`
5	$[Z, X, Y]$	$\left[\frac{1}{4}, x, 0\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}-x, 0\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{3}{4}, \frac{1}{2}-x, 0\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, x, 0\right]$	`[8]`
9	$[Y, Z, X]$	$\left[0, \frac{1}{4}, x\right]$	`[9]`

continued ...

Table 21

No.	vector	center	mapping
10	$[-Y, Z, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}-x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[0, \frac{1}{4}, x\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, x+\frac{1}{4}, 0\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}-x, \frac{1}{2}\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{3}{4}, \frac{3}{4}-x, \frac{1}{2}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{3}{4}, x+\frac{1}{4}, 0\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{4}-x, 0, \frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{4}-x, 0, \frac{1}{4}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[0, \frac{1}{4}, \frac{1}{4}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{4}, x+\frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{4}, x+\frac{3}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[0, \frac{1}{4}, \frac{1}{4}-x\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[-x, 0, \frac{3}{4}\right]$	[25]
26	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, 0, \frac{1}{4}\right]$	[26]
27	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, 0, \frac{1}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[-x, 0, \frac{3}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{3}{4}, -x, 0\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{4}, x+\frac{1}{2}, 0\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{1}{4}, x+\frac{1}{2}, 0\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{3}{4}, -x, 0\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[0, \frac{3}{4}, -x\right]$	[33]
34	$[Y, -Z, X]$	$\left[0, \frac{1}{4}, x+\frac{1}{2}\right]$	[34]
35	$[-Y, Z, X]$	$\left[0, \frac{1}{4}, x+\frac{1}{2}\right]$	[35]
36	$[Y, Z, -X]$	$\left[0, \frac{3}{4}, -x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}-x, 0\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{4}, x+\frac{1}{4}, \frac{1}{2}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{4}, x+\frac{1}{4}, \frac{1}{2}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}-x, 0\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{4}-x, \frac{1}{2}, \frac{3}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x+\frac{3}{4}, 0, \frac{3}{4}\right]$	[42]
43	$[X, Z, Y]$	$\left[x+\frac{3}{4}, 0, \frac{3}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{4}-x, \frac{1}{2}, \frac{3}{4}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[0, \frac{3}{4}, x+\frac{3}{4}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-x\right]$	[47]
48	$[Z, Y, X]$	$\left[0, \frac{3}{4}, x+\frac{3}{4}\right]$	[48]

* Wyckoff site: 48g, site symmetry: ..2

Table 22: Wyckoff bond: 48a@48g, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{8}, y, \frac{1}{4}-y\right]$	[1,-19]
2	$[-Y, -X, X]$	$\left[\frac{3}{8}, -y, \frac{3}{4}-y\right]$	[2,-17]

continued ...

Table 22

No.	vector	center	mapping
3	$[-Y, X, -X]$	$\left[\frac{7}{8}, y+\frac{1}{2}, y+\frac{1}{4}\right]$	[3,-20]
4	$[Y, -X, -X]$	$\left[\frac{5}{8}, \frac{1}{2}-y, y+\frac{3}{4}\right]$	[4,-18]
5	$[X, Y, X]$	$\left[\frac{1}{4}-y, \frac{1}{8}, y\right]$	[5,-14]
6	$[X, -Y, -X]$	$\left[\frac{3}{4}-y, \frac{3}{8}, -y\right]$	[6,-16]
7	$[-X, -Y, X]$	$\left[y+\frac{1}{4}, \frac{7}{8}, y+\frac{1}{2}\right]$	[7,-13]
8	$[-X, Y, -X]$	$\left[y+\frac{3}{4}, \frac{5}{8}, \frac{1}{2}-y\right]$	[8,-15]
9	$[X, X, Y]$	$\left[y, \frac{1}{4}-y, \frac{1}{8}\right]$	[9,-24]
10	$[-X, X, -Y]$	$\left[-y, \frac{3}{4}-y, \frac{3}{8}\right]$	[10,-22]
11	$[X, -X, -Y]$	$\left[y+\frac{1}{2}, y+\frac{1}{4}, \frac{7}{8}\right]$	[11,-23]
12	$[-X, -X, Y]$	$\left[\frac{1}{2}-y, y+\frac{3}{4}, \frac{5}{8}\right]$	[12,-21]
13	$[-Y, -X, -X]$	$\left[\frac{7}{8}, -y, y+\frac{3}{4}\right]$	[25,-43]
14	$[Y, X, -X]$	$\left[\frac{5}{8}, y, y+\frac{1}{4}\right]$	[26,-41]
15	$[Y, -X, X]$	$\left[\frac{1}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$	[27,-44]
16	$[-Y, X, X]$	$\left[\frac{3}{8}, y+\frac{1}{2}, \frac{1}{4}-y\right]$	[28,-42]
17	$[-X, -Y, -X]$	$\left[y+\frac{3}{4}, \frac{7}{8}, -y\right]$	[29,-38]
18	$[-X, Y, X]$	$\left[y+\frac{1}{4}, \frac{5}{8}, y\right]$	[30,-40]
19	$[X, Y, -X]$	$\left[\frac{3}{4}-y, \frac{1}{8}, \frac{1}{2}-y\right]$	[31,-37]
20	$[X, -Y, X]$	$\left[\frac{1}{4}-y, \frac{3}{8}, y+\frac{1}{2}\right]$	[32,-39]
21	$[-X, -X, -Y]$	$\left[-y, y+\frac{3}{4}, \frac{7}{8}\right]$	[33,-48]
22	$[X, -X, Y]$	$\left[y, y+\frac{1}{4}, \frac{5}{8}\right]$	[34,-46]
23	$[-X, X, Y]$	$\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{1}{8}\right]$	[35,-47]
24	$[X, X, -Y]$	$\left[y+\frac{1}{2}, \frac{1}{4}-y, \frac{3}{8}\right]$	[36,-45]

Table 23: Wyckoff bond: 48b@48g, bond symmetry: ..2

No.	vector	center	mapping
1	$[0, X, -X]$	$\left[\frac{1}{8}, y, \frac{1}{4}-y\right]$	[1,19]
2	$[0, -X, -X]$	$\left[\frac{3}{8}, -y, \frac{3}{4}-y\right]$	[2,17]
3	$[0, X, X]$	$\left[\frac{7}{8}, y+\frac{1}{2}, y+\frac{1}{4}\right]$	[3,20]
4	$[0, -X, X]$	$\left[\frac{5}{8}, \frac{1}{2}-y, y+\frac{3}{4}\right]$	[4,18]
5	$[-X, 0, X]$	$\left[\frac{1}{4}-y, \frac{1}{8}, y\right]$	[5,14]
6	$[-X, 0, -X]$	$\left[\frac{3}{4}-y, \frac{3}{8}, -y\right]$	[6,16]
7	$[X, 0, X]$	$\left[y+\frac{1}{4}, \frac{7}{8}, y+\frac{1}{2}\right]$	[7,13]
8	$[X, 0, -X]$	$\left[y+\frac{3}{4}, \frac{5}{8}, \frac{1}{2}-y\right]$	[8,15]
9	$[X, -X, 0]$	$\left[y, \frac{1}{4}-y, \frac{1}{8}\right]$	[9,24]
10	$[-X, -X, 0]$	$\left[-y, \frac{3}{4}-y, \frac{3}{8}\right]$	[10,22]
11	$[X, X, 0]$	$\left[y+\frac{1}{2}, y+\frac{1}{4}, \frac{7}{8}\right]$	[11,23]
12	$[-X, X, 0]$	$\left[\frac{1}{2}-y, y+\frac{3}{4}, \frac{5}{8}\right]$	[12,21]
13	$[0, -X, X]$	$\left[\frac{7}{8}, -y, y+\frac{3}{4}\right]$	[25,43]
14	$[0, X, X]$	$\left[\frac{5}{8}, y, y+\frac{1}{4}\right]$	[26,41]
15	$[0, -X, -X]$	$\left[\frac{1}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$	[27,44]
16	$[0, X, -X]$	$\left[\frac{3}{8}, y+\frac{1}{2}, \frac{1}{4}-y\right]$	[28,42]
17	$[X, 0, -X]$	$\left[y+\frac{3}{4}, \frac{7}{8}, -y\right]$	[29,38]
18	$[X, 0, X]$	$\left[y+\frac{1}{4}, \frac{5}{8}, y\right]$	[30,40]
19	$[-X, 0, -X]$	$\left[\frac{3}{4}-y, \frac{1}{8}, \frac{1}{2}-y\right]$	[31,37]

continued ...

Table 23

No.	vector	center	mapping
20	$[-X,\ 0,\ X]$	$\left[\frac{1}{4}-y,\ \frac{3}{8},\ y+\frac{1}{2}\right]$	[32,39]
21	$[-X,\ X,\ 0]$	$\left[-y,\ y+\frac{3}{4},\ \frac{7}{8}\right]$	[33,48]
22	$[X,\ X,\ 0]$	$\left[y,\ y+\frac{1}{4},\ \frac{5}{8}\right]$	[34,46]
23	$[-X,\ -X,\ 0]$	$\left[\frac{1}{2}-y,\ \frac{3}{4}-y,\ \frac{1}{8}\right]$	[35,47]
24	$[X,\ -X,\ 0]$	$\left[y+\frac{1}{2},\ \frac{1}{4}-y,\ \frac{3}{8}\right]$	[36,45]

Table 24: Wyckoff bond: 96c@48g, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{8},\ y,\ \frac{1}{4}-y\right]$	[1]
2	$[-X,\ -Y,\ Z]$	$\left[\frac{3}{8},\ -y,\ \frac{3}{4}-y\right]$	[2]
3	$[-X,\ Y,\ -Z]$	$\left[\frac{7}{8},\ y+\frac{1}{2},\ y+\frac{1}{4}\right]$	[3]
4	$[X,\ -Y,\ -Z]$	$\left[\frac{5}{8},\ \frac{1}{2}-y,\ y+\frac{3}{4}\right]$	[4]
5	$[Z,\ X,\ Y]$	$\left[\frac{1}{4}-y,\ \frac{1}{8},\ y\right]$	[5]
6	$[Z,\ -X,\ -Y]$	$\left[\frac{3}{4}-y,\ \frac{3}{8},\ -y\right]$	[6]
7	$[-Z,\ -X,\ Y]$	$\left[y+\frac{1}{4},\ \frac{7}{8},\ y+\frac{1}{2}\right]$	[7]
8	$[-Z,\ X,\ -Y]$	$\left[y+\frac{3}{4},\ \frac{5}{8},\ \frac{1}{2}-y\right]$	[8]
9	$[Y,\ Z,\ X]$	$\left[y,\ \frac{1}{4}-y,\ \frac{1}{8}\right]$	[9]
10	$[-Y,\ Z,\ -X]$	$\left[-y,\ \frac{3}{4}-y,\ \frac{3}{8}\right]$	[10]
11	$[Y,\ -Z,\ -X]$	$\left[y+\frac{1}{2},\ y+\frac{1}{4},\ \frac{7}{8}\right]$	[11]
12	$[-Y,\ -Z,\ X]$	$\left[\frac{1}{2}-y,\ y+\frac{3}{4},\ \frac{5}{8}\right]$	[12]
13	$[Y,\ X,\ -Z]$	$\left[y+\frac{1}{4},\ \frac{7}{8},\ y+\frac{1}{2}\right]$	[13]
14	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{4}-y,\ \frac{1}{8},\ y\right]$	[14]
15	$[Y,\ -X,\ Z]$	$\left[y+\frac{3}{4},\ \frac{5}{8},\ \frac{1}{2}-y\right]$	[15]
16	$[-Y,\ X,\ Z]$	$\left[\frac{3}{4}-y,\ \frac{3}{8},\ -y\right]$	[16]
17	$[X,\ Z,\ -Y]$	$\left[\frac{3}{8},\ -y,\ \frac{3}{4}-y\right]$	[17]
18	$[-X,\ Z,\ Y]$	$\left[\frac{5}{8},\ \frac{1}{2}-y,\ y+\frac{3}{4}\right]$	[18]
19	$[-X,\ -Z,\ -Y]$	$\left[\frac{1}{8},\ y,\ \frac{1}{4}-y\right]$	[19]
20	$[X,\ -Z,\ Y]$	$\left[\frac{7}{8},\ y+\frac{1}{2},\ y+\frac{1}{4}\right]$	[20]
21	$[Z,\ Y,\ -X]$	$\left[\frac{1}{2}-y,\ y+\frac{3}{4},\ \frac{5}{8}\right]$	[21]
22	$[Z,\ -Y,\ X]$	$\left[-y,\ \frac{3}{4}-y,\ \frac{3}{8}\right]$	[22]
23	$[-Z,\ Y,\ X]$	$\left[y+\frac{1}{2},\ y+\frac{1}{4},\ \frac{7}{8}\right]$	[23]
24	$[-Z,\ -Y,\ -X]$	$\left[y,\ \frac{1}{4}-y,\ \frac{1}{8}\right]$	[24]
25	$[-X,\ -Y,\ -Z]$	$\left[\frac{7}{8},\ -y,\ y+\frac{3}{4}\right]$	[25]
26	$[X,\ Y,\ -Z]$	$\left[\frac{5}{8},\ y,\ y+\frac{1}{4}\right]$	[26]
27	$[X,\ -Y,\ Z]$	$\left[\frac{1}{8},\ \frac{1}{2}-y,\ \frac{3}{4}-y\right]$	[27]
28	$[-X,\ Y,\ Z]$	$\left[\frac{3}{8},\ y+\frac{1}{2},\ \frac{1}{4}-y\right]$	[28]
29	$[-Z,\ -X,\ -Y]$	$\left[y+\frac{3}{4},\ \frac{7}{8},\ -y\right]$	[29]
30	$[-Z,\ X,\ Y]$	$\left[y+\frac{1}{4},\ \frac{5}{8},\ y\right]$	[30]
31	$[Z,\ X,\ -Y]$	$\left[\frac{3}{4}-y,\ \frac{1}{8},\ \frac{1}{2}-y\right]$	[31]
32	$[Z,\ -X,\ Y]$	$\left[\frac{1}{4}-y,\ \frac{3}{8},\ y+\frac{1}{2}\right]$	[32]
33	$[-Y,\ -Z,\ -X]$	$\left[-y,\ y+\frac{3}{4},\ \frac{7}{8}\right]$	[33]
34	$[Y,\ -Z,\ X]$	$\left[y,\ y+\frac{1}{4},\ \frac{5}{8}\right]$	[34]
35	$[-Y,\ Z,\ X]$	$\left[\frac{1}{2}-y,\ \frac{3}{4}-y,\ \frac{1}{8}\right]$	[35]
36	$[Y,\ Z,\ -X]$	$\left[y+\frac{1}{2},\ \frac{1}{4}-y,\ \frac{3}{8}\right]$	[36]

continued ...

Table 24

No.	vector	center	mapping
37	$[-Y, -X, Z]$	$\left[\frac{3}{4} - y, \frac{1}{8}, \frac{1}{2} - y\right]$	[37]
38	$[Y, X, Z]$	$\left[y + \frac{3}{4}, \frac{7}{8}, -y\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{4} - y, \frac{3}{8}, y + \frac{1}{2}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[y + \frac{1}{4}, \frac{5}{8}, y\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{5}{8}, y, y + \frac{1}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{3}{8}, y + \frac{1}{2}, \frac{1}{4} - y\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{7}{8}, -y, y + \frac{3}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{8}, \frac{1}{2} - y, \frac{3}{4} - y\right]$	[44]
45	$[-Z, -Y, X]$	$\left[y + \frac{1}{2}, \frac{1}{4} - y, \frac{3}{8}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[y, y + \frac{1}{4}, \frac{5}{8}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{1}{2} - y, \frac{3}{4} - y, \frac{1}{8}\right]$	[47]
48	$[Z, Y, X]$	$\left[-y, y + \frac{3}{4}, \frac{7}{8}\right]$	[48]

* Wyckoff site: 96h, site symmetry: 1

Table 25: Wyckoff bond: 96a@96h, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, -y, z + \frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, y + \frac{1}{2}, \frac{1}{2} - z\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - y, -z\right]$	[4]
5	$[Z, X, Y]$	$[z, x, y]$	[5]
6	$[Z, -X, -Y]$	$\left[z + \frac{1}{2}, \frac{1}{2} - x, -y\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2} - z, -x, y + \frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[-z, x + \frac{1}{2}, \frac{1}{2} - y\right]$	[8]
9	$[Y, Z, X]$	$[y, z, x]$	[9]
10	$[-Y, Z, -X]$	$\left[-y, z + \frac{1}{2}, \frac{1}{2} - x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[y + \frac{1}{2}, \frac{1}{2} - z, -x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2} - y, -z, x + \frac{1}{2}\right]$	[12]
13	$[Y, X, -Z]$	$\left[y + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{4} - z\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4} - y, \frac{3}{4} - x, \frac{3}{4} - z\right]$	[14]
15	$[Y, -X, Z]$	$\left[y + \frac{1}{4}, \frac{1}{4} - x, z + \frac{3}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{4} - y, x + \frac{3}{4}, z + \frac{1}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x + \frac{3}{4}, z + \frac{1}{4}, \frac{1}{4} - y\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{4} - x, z + \frac{3}{4}, y + \frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{3}{4} - x, \frac{3}{4} - z, \frac{3}{4} - y\right]$	[19]
20	$[X, -Z, Y]$	$\left[x + \frac{1}{4}, \frac{1}{4} - z, y + \frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[z + \frac{3}{4}, y + \frac{1}{4}, \frac{1}{4} - x\right]$	[21]
22	$[Z, -Y, X]$	$\left[z + \frac{1}{4}, \frac{1}{4} - y, x + \frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{4} - z, y + \frac{3}{4}, x + \frac{1}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{3}{4} - z, \frac{3}{4} - y, \frac{3}{4} - x\right]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[25]
26	$[X, Y, -Z]$	$\left[x + \frac{1}{2}, y, \frac{1}{2} - z\right]$	[26]
27	$[X, -Y, Z]$	$\left[x, \frac{1}{2} - y, z + \frac{1}{2}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{1}{2} - x, y + \frac{1}{2}, z\right]$	[28]
29	$[-Z, -X, -Y]$	$[-z, -x, -y]$	[29]

continued ...

Table 25

No.	vector	center	mapping
30	$[-Z,\, X,\, Y]$	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, y\right]$	[30]
31	$[Z,\, X,\, -Y]$	$\left[z+\frac{1}{2},\, x,\, \frac{1}{2}-y\right]$	[31]
32	$[Z,\, -X,\, Y]$	$\left[z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[32]
33	$[-Y,\, -Z,\, -X]$	$[-y,\, -z,\, -x]$	[33]
34	$[Y,\, -Z,\, X]$	$\left[y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[34]
35	$[-Y,\, Z,\, X]$	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, x\right]$	[35]
36	$[Y,\, Z,\, -X]$	$\left[y+\frac{1}{2},\, z,\, \frac{1}{2}-x\right]$	[36]
37	$[-Y,\, -X,\, Z]$	$\left[\frac{1}{4}-y,\, \frac{3}{4}-x,\, z+\frac{3}{4}\right]$	[37]
38	$[Y,\, X,\, Z]$	$\left[y+\frac{1}{4},\, x+\frac{1}{4},\, z+\frac{1}{4}\right]$	[38]
39	$[-Y,\, X,\, -Z]$	$\left[\frac{3}{4}-y,\, x+\frac{3}{4},\, \frac{1}{4}-z\right]$	[39]
40	$[Y,\, -X,\, -Z]$	$\left[y+\frac{3}{4},\, \frac{1}{4}-x,\, \frac{3}{4}-z\right]$	[40]
41	$[-X,\, -Z,\, Y]$	$\left[\frac{1}{4}-x,\, \frac{3}{4}-z,\, y+\frac{3}{4}\right]$	[41]
42	$[X,\, -Z,\, -Y]$	$\left[x+\frac{3}{4},\, \frac{1}{4}-z,\, \frac{3}{4}-y\right]$	[42]
43	$[X,\, Z,\, Y]$	$\left[x+\frac{1}{4},\, z+\frac{1}{4},\, y+\frac{1}{4}\right]$	[43]
44	$[-X,\, Z,\, -Y]$	$\left[\frac{3}{4}-x,\, z+\frac{3}{4},\, \frac{1}{4}-y\right]$	[44]
45	$[-Z,\, -Y,\, X]$	$\left[\frac{1}{4}-z,\, \frac{3}{4}-y,\, x+\frac{3}{4}\right]$	[45]
46	$[-Z,\, Y,\, -X]$	$\left[\frac{3}{4}-z,\, y+\frac{3}{4},\, \frac{1}{4}-x\right]$	[46]
47	$[Z,\, -Y,\, -X]$	$\left[z+\frac{3}{4},\, \frac{1}{4}-y,\, \frac{3}{4}-x\right]$	[47]
48	$[Z,\, Y,\, X]$	$\left[z+\frac{1}{4},\, y+\frac{1}{4},\, x+\frac{1}{4}\right]$	[48]